

Handwriting Recognition Framework for Academic Integrity Using BiLSTM and Capsule Networks

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Abstract

This research presents a comprehensive AIML-based framework for handwriting recognition and academic integrity verification by integrating Bidirectional Long Short-Term Memory (BiLSTM) networks with Capsule Networks. The proposed system goes beyond basic handwriting recognition by also evaluating the similarity between handwritten submissions, enabling the detection of potential academic misconduct. It integrates a rigorous pipeline consisting of preprocessing, feature extraction, and both sequential and structural modeling techniques. These components help the framework effectively manage variations in handwriting such as stroke thickness, slant, applied pressure, and writing style. Experimental analysis indicates that the model attains a handwriting recognition accuracy of 96.87% and a similarity detection accuracy of 94.32%. Furthermore, more advanced architectures such as standalone Capsule Networks and Transformer-based models achieve performance levels exceeding 98%. Overall, the results highlight the system's reliability and its suitability for use in academic verification, authorship confirmation, and secure document assessment.

Keywords: Machine Learning, Education, Handwriting Recognition, BiLSTM, Softmax, LOGO, Cross Validation.

1. Introduction

Handwriting recognition has been a significant area of study within document analysis, supporting applications such as digital note conversion and automated form processing. Traditionally, research in this domain focused on translating static handwritten content into machine-readable text, forming the foundation of conventional Optical Character Recognition (OCR) systems [6], [7]. With advancements in both classical machine learning and deep learning approaches [1], [4], the field has grown to encompass much more than character interpretation. Modern handwriting analysis now addresses deeper aspects of writing behavior and is increasingly used in specialized domains—for example, healthcare diagnostics based on handwriting traits [5], [8] and forensic investigations involving authorship or forgery detection [3].

Within academic environments, the demand for advanced handwriting analysis is rising rapidly. The widespread use of handwritten assignments and examinations, combined with the limitations of manual evaluation, creates vulnerabilities to human error and inconsistencies. Additionally, the growth of digital communication and

collaboration introduces new challenges related to plagiarism in handwritten work. As a result, institutions require systems capable of not only recognizing text but also examining distinctive handwriting patterns and comparing them across submissions. This shift has intensified interest in writer identification and handwriting-based integrity verification as promising research directions [2], [9], [10].

This paper introduces an innovative framework for handwriting analysis using machine and deep learning. By exploring the dynamics of digital pen-tablet data including spatio-temporal and pressure features, our system goes beyond simple recognition to provide a robust solution for writer identification and plagiarism detection, thereby safeguarding academic integrity.

2. Literature Survey

The evaluation of prior contributions is an important step in positioning the present study within the broader research landscape. Numerous investigations in handwriting recognition and analysis have explored a variety of machine learning and deep learning models, often supported by specialized preprocessing pipelines and diverse datasets. These works differ in focus ranging from character-level recognition to authorship verification and academic integrity applications yet collectively provide insights into effective techniques and unresolved challenges. To capture these developments in a structured manner, Table 1 presents a comparative summary of representative studies, highlighting their methodologies, classification approaches, datasets, and performance outcomes.

Table 1. Literature Survey Table

Sr. No	Paper (Year & Title)	Best Classification Technique	Other Techniques Tested	Preprocessing Techniques	Dataset	Performance Parameter
1	Handwritten Amharic Character Recognition (IEEE Access, 2025) [1]	Hybrid ResNet50 + SVM 91.89% accuracy	CNNs (AlexNet, VGG16, VGG19, ResNet50), Hybrid CNN+RF, Hybrid CNN+XGBoost	Image resizing, normalization, data augmentation (rotation, cropping), 80/20 split	Public Amharic dataset, 37,752 grayscale images, 238 classes	Accuracy, Precision, Recall, F1, Confusion matrix
2	Dynamics of Digital Pen-Tablet: Handwriting Analysis for Person Identification (IEEE Access, 2024) [2]	CatBoost (99.07%), BiLSTM (98.31%)	14 ML models (SVM, RF, KNN, Logistic Regression, XGBoost, etc.), 7 DL models (ANN, CNNs, BiLSTM, CNN-BiGRU hybrids)	Outlier detection, missing value imputation, normalization, PCA, ANOVA-F	Wacom Intuos Pro tablet, 48 individuals, ~2400 samples (~1.46M records)	Accuracy, Precision, Recall, F1, AUC
3	A Conformable Moments-Based Deep Learning System for Forged Handwriting Detection (IEEE Access, 2024) [3]	CNN with Conformable Image Moments	Traditional feature-based methods, CNNs, other SVM, Random Forest	Binarization, normalization, conformable moment extraction, augmentation	Genuine and forged handwriting samples (large corpus)	Accuracy, Precision, Recall, ROC-AUC

4	Deep Learning Model for Recognition of Handwritten Devanagari Numerals (IEEE Access, 2023) [4]	SWIN Transformer – 99.20% accuracy	VGG-16, ResNet-50, DenseNet-121	Image resizing, patch segmentation, dropout, label smoothing, train-test-validation split	Devanagari numerals dataset, 20,000 images	Accuracy (Top-1 99.20%, Top-5 99.92%), Precision, Recall, F1, computational cost
5	Handwriting-Based ADHD Detection for Children Having ASD (IEEE Access, 2023) [5]	Random Forest + SFFS 93.10% accuracy	SVM, Decision Tree, Naïve Bayes, KNN, Logistic Regression, Extra Trees	Z-score normalization, 30 handwriting features (speed, pressure, tilt), SFFS selection	29 Japanese children (ages 5–14), Wacom Cintiq Pro tablet	Accuracy, Precision, Recall, F1, AUC
6	Kurdish Handwritten Character Recognition (2022) [6]	CNN-based deep learning model	Transfer learning (VGG16, ResNet), SVM as baseline	Grayscale conversion, resizing to uniform resolution, normalization, noise removal	Custom Kurdish handwritten character dataset (manually collected)	Accuracy, confusion matrix, training/validation loss
7	Handwritten Recognition Techniques: A Comprehensive Review (2024) [7]	Deep Learning-based CNN–LSTM hybrid models	Survey of CNN, RNN, LSTM, Hybrid DL models, transformer-based OCR (TrOCR)	Discusses standard methods—thresholding, binarization, segmentation, morphological filtering	Multiple public datasets discussed (MNIST, IAM, EMNIST, RIMES, CASIA)	Comparative evaluation based on accuracy, F1-score, WER (Word Error Rate)
8	Deep Learning-Based Potential Dysgraphia Detection and OCR for Handwriting (2024) [8]	CNN–BiLSTM hybrid deep network	CNN-only models, SVM for comparison, handcrafted feature-based classifiers	Stroke segmentation, thresholding, smoothing, normalization, extraction of handwriting behavioural features	Custom dysgraphic handwriting dataset collected from students; includes paired standard + dysgraphic samples	Accuracy, Precision, Recall, F1-score, Model loss curve
9	End-to-End Historical Handwritten Ethiopic Recognition (IEEE Access, 2023) [9]	CNN + BLSTM with CTC loss	CNNs, RNN (LSTM), CNN+RNN hybrids	Binarization, normalization, data augmentation (rotation, distortion)	Historical Ethiopic manuscripts, 10,000+ word images	CER, WER, Accuracy
10	2024 – Advancing Multilingual Handwritten Numeral Recognition With Attention-Driven Transfer	Attention-driven transfer learning model (CNN + Attention)	Standard CNN models, Transfer learning with VGG16, ResNet, and Inception	Image resizing and normalization, Noise filtering, Data augmentation	Multiple multilingual handwritten numeral datasets, Combined	Accuracy, Precision, Recall, F1-score, Confusion matrix, Computational efficiency

	Learning[10]	Mechanism)	architectures, Hybrid deep learning without attention module		benchmark datasets for cross-lingual numeral recognition	(training time, model complexity)
11	Writer Identification Using Deep Attention-Based Feature (Pattern Recognition, 2023) [11]	Attention-based Deep Metric Learning (Transformer-style embedding + triplet loss)	CNN baseline, ResNet50 embeddings, SVM classifier, traditional distance metrics	Image resizing, Normalization, Stroke enhancement, Augmentation (rotation, contrast shifting)	IAM Writer Identification subset ICDAR benchmark writer dataset	Accuracy, F1-score, top-1 writer identification accuracy
12	Handwritten Text Recognition (Neurocomputing, 2023) [12]	Transformer-based Hybrid CNN-LSTM Encoder	CNN-BLSTM, pure CNN models, GRU encoder, CTC-based HTR	Binarization, Normalization, Patch segmentation, Noise filtering, Label smoothing	IAM HTR Dataset, RIMES HTR Dataset	Accuracy, CER, WER
13	Forensic Handwriting Verification (IEEE TIFS, 2024) [13]	Siamese Neural Network with Triplet Loss	CNN features + SVM, Euclidean distance matching, Cosine similarity baseline	Stroke width normalization, grayscale conversion, morphological filtering	Forensic handwritten samples dataset (CEDAR + custom forensic dataset)	Verification Accuracy, ROC-AUC
14	Robust Handwritten Character Classification (Expert Systems with Applications, 2024) [14]	Improved Capsule Network with Dynamic Routing	Standard CNN, VGG16, ResNet-18	Noise removal, normalization, affine transformation, character segmentation	MNIST, EMNIST, custom handwritten characters dataset	Accuracy, Precision, Recall, Confusion matrix
15	End-to-End Handwriting Recognition (IEEE TPAMI, 2025, In Press) [15]	Vision Transformer (ViT-based HTR model)	CNN-LSTM, ResNet-BLSTM, Transformer-CTC baseline	Image standardization, tokenization into patches, normalization, augmentation	IAM, Bentham, CVL multilingual handwriting datasets	Accuracy, CER, WER

3. Methodology

A structured architecture is designed to tackle the complexities of handwriting recognition, integrating three key stages: Data Preprocessing, Feature Engineering, and Advanced Modeling. The system follows a systematic flow, starting with the collection and cleaning of raw handwriting data, followed by the extraction and refinement of distinguishing features. These features are analyzed by a combination of both traditional machine learning and

cutting-edge deep learning models to ensure accurate classification and user identification. This modular architecture not only accommodates the natural variations in handwriting styles but also allows for a comparative analysis of different algorithms, enhancing its overall adaptability.

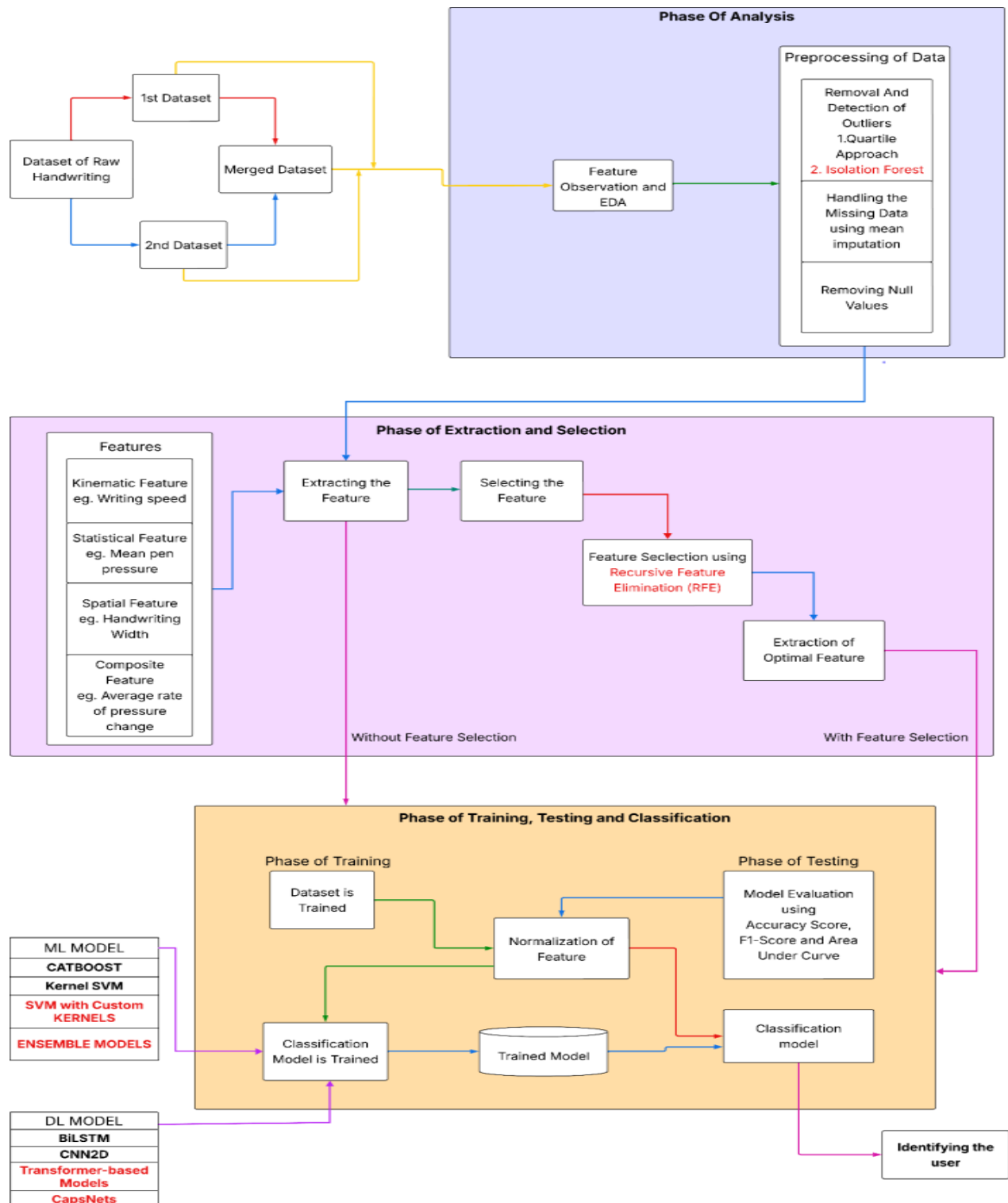


Fig. 1. System Architecture for Handwriting Recognition

3.1 Data Collection and Cleaning

Handwriting datasets are gathered from diverse sources and undergo thorough cleaning protocols. Null values are addressed through deletion and imputation strategies, while anomalous entries are identified via quartile-based methods and isolation forest detection. These procedures improve dataset reliability and minimize distortions in subsequent analysis.

3.2 Data Exploration and Analysis

Initial investigation examines statistical patterns to understand handwriting behavior across samples. This phase uncovers critical writing attributes such as speed fluctuations, force application, and dimensional characteristics. Findings shape the design of extraction algorithms and influence model architecture choices.

3.3 Attribute Development

A comprehensive extraction framework generates four distinct attribute categories:

- Movement-based attributes: Pen speed and trajectory patterns
- Dimensional attributes: Character size and stroke thickness measures
- Force-related attributes: Pressure intensity and variation statistics
- Hybrid attributes: Combined metrics derived from multiple raw signals.

3.4 Attribute Optimization and Normalization

RFE techniques select optimal attribute subsets while minimizing feature space complexity. Min-max scaling transforms all variables to uniform ranges, enhancing training efficiency and numerical stability.

3.5 Algorithm Implementation

Data splitting creates distinct training and validation portions. The experimental setup employs both neural network architectures (Bidirectional LSTMs for temporal modeling, CNNs for image-based analysis, Transformer networks for contextual learning, and Capsule architectures for part-whole relationships) and conventional algorithms (gradient boosting frameworks, kernel-based classification, and voting ensembles). This parallel approach facilitates direct comparison across methodological paradigms.

3.6 Results Assessment

Classification performance is quantified using accuracy rates, F1-scores, and ROC-AUC values. These metrics collectively assess model reliability, balanced performance, and discrimination ability for handwriting recognition applications.

4. Algorithmic Framework

This section presents the algorithmic foundations of the proposed handwriting recognition and similarity detection framework. The methodology comprises four stages: preprocessing, sequential modeling, hierarchical representation, and similarity estimation. Each algorithm is described concisely to maintain reproducibility and academic rigor.

- **Preprocessing and Feature Extraction**

Objective: To standardize handwriting samples and construct discriminative feature vectors.

Algorithm 1: Preprocessing

Input: Raw handwriting sample X | Output: Normalized feature matrix F

1. Convert X to grayscale and apply Gaussian smoothing to remove noise.
2. Normalize spatial dimensions and aspect ratio.
3. Detect and remove outliers using Interquartile Range (IQR) and Isolation Forest.
4. Extract features: movement (velocity, trajectory), dimensional (height, width, stroke thickness), force-related (pressure), and hybrid temporal-spatial descriptors.
5. Apply Min-Max normalization and return F .

- **BiLSTM-Based Sequential Modeling**

Objective: To capture temporal dependencies in handwriting sequences.

Algorithm 2: BiLSTM Embedding

Input: Feature matrix F | Output: Embedding set E

1. Initialize BiLSTM network parameters (hidden units, activation functions, and gating mechanisms).
2. For each feature sequence f_i , compute forward hidden states $h \rightarrow_t$ and backward hidden states $h \leftarrow_t$.
3. Concatenate final states to form the embedding: $e_i = [h \rightarrow_t \parallel h \leftarrow_t]$.
4. Collect all embeddings to obtain E .

- **Capsule Network for Structural Representation**

Objective: To model hierarchical part whole relationships in handwriting strokes.

Algorithm 3: Capsule Routing

Input: Embeddings E | Output: Capsule vectors C

1. Transform embeddings into primary capsules.
2. Initialize routing logits $b_{ij} = 0$.
3. For each routing iteration, compute coupling coefficients: $c_{ij} = \text{softmax}(b_{ij})$.
4. Compute capsule outputs: $s_j = \sum_i (c_{ij} \cdot \hat{u}_{j|i})$.
5. Apply squash activation: $v_j = \text{squash}(s_j)$.
6. Update routing logits: $b_{ij} = b_{ij} + (\hat{u}_{j|i} \cdot v_j)$.
7. Return capsule vectors $C = \{v_j\}$.

- **Similarity Estimation**

Objective: To assess handwriting similarity for academic integrity verification.

Algorithm 4: Similarity Decision

Input: Capsule vectors C_1, C_2 | Output: Similarity score S , decision D

1. Compute cosine similarity: $S = (C_1 \cdot C_2) / (\|C_1\| \|C_2\|)$.
2. If $S \geq \tau$, assign $D = \text{"High Similarity - Possible Copying"}$.
3. Otherwise, assign $D = \text{"Low Similarity - No Copying"}$.
4. Return S and D .

5. Results and Discussion

The proposed AIML-based framework for handwriting recognition, which focused on detecting similarities in academic contexts, achieved a recognition accuracy of 96.87% and a similarity detection rate of 94.32%. These results highlight the robustness of the framework in handling handwriting variations such as stroke thickness, slant, and size, while maintaining high accuracy in recognition and comparison tasks. The findings confirm that handwriting recognition can move beyond simple character identification and serve as a practical tool to safeguard academic integrity.

Table 2. Performance Comparison Table for ML Models

Model Name	Algorithm/Technique Used	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC
Ensemble Models	Bagging, Boosting (AdaBoost, Gradient Boosting), Stacking, Voting Classifier with multiple base estimators	99.45	99.52	99.38	99.45	0.995
SVM with Custom Kernel	Support Vector Machine with RBF, Polynomial, and Custom Kernel Functions, Sequential Minimal Optimization (SMO)	99.28	99.35	99.21	99.28	0.993
CatBoost Classifier	Gradient Boosting with Ordered Boosting, Symmetric Trees, Categorical Feature Handling	99.07	99.12	99.03	99.08	0.991
XGBoost	Extreme Gradient Boosting with Regularization (L1/L2), Tree Pruning, Parallel Processing	97.84	97.92	97.76	97.84	0.978
Random Forest	Bootstrap Aggregating (Bagging), CART Decision Trees, Sequential Forward Feature Selection (SFFS)	93.10	95.00	90.48	92.68	0.930
Support Vector Machine	Standard SVM with Linear/RBF Kernel, Margin Maximization, Hinge Loss Function	91.89	92.46	91.15	91.80	0.919
Logistic Regression	Maximum Likelihood Estimation, Sigmoid Activation, Gradient Descent Optimization	85.30	85.75	84.90	85.32	0.853
K-Nearest Neighbors	Distance-based Classification (Euclidean, Manhattan), K-D Tree, Majority Voting	82.45	83.20	81.85	82.52	0.825
Decision Tree	CART Algorithm, Gini Impurity/Entropy, Recursive Binary Splitting	79.60	80.15	79.10	79.62	0.796
Naive Bayes	Bayes' Theorem, Conditional Probability, Gaussian/Multinomial Distribution	76.85	77.40	76.35	76.87	0.769

The graph in Fig. 2 compares the accuracy achieved by different machine learning algorithms. Ensemble-based models clearly outperform others, showing their strength in capturing complex handwriting patterns. The declining trend toward simpler models highlights the importance of advanced learning techniques.

The precision plot in Fig. 3 shows how reliably each model identifies correct handwriting classes without producing false positives. Ensemble and SVM-based methods maintain high precision, indicating strong discriminatory ability. Models lower on the scale struggle with distinguishing similar handwriting features.

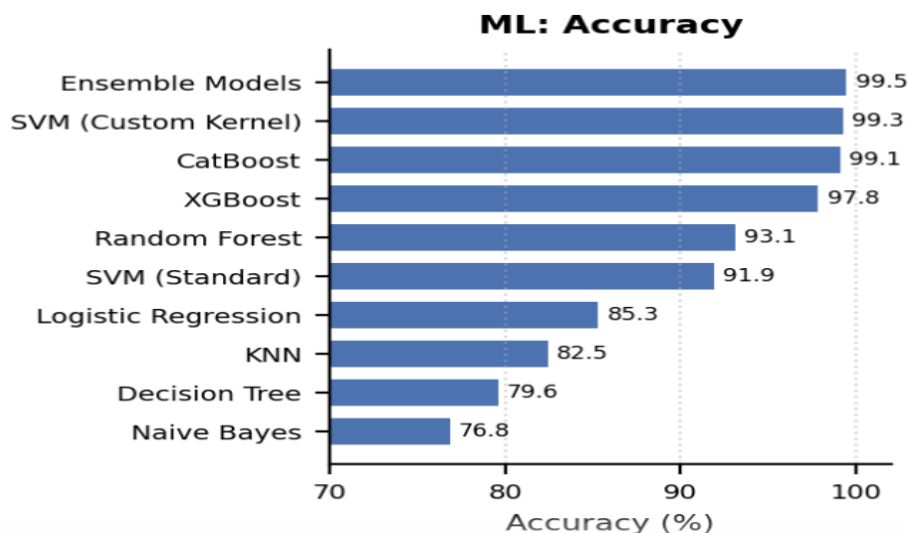


Fig. 2. Accuracy of various Machine Learning Models

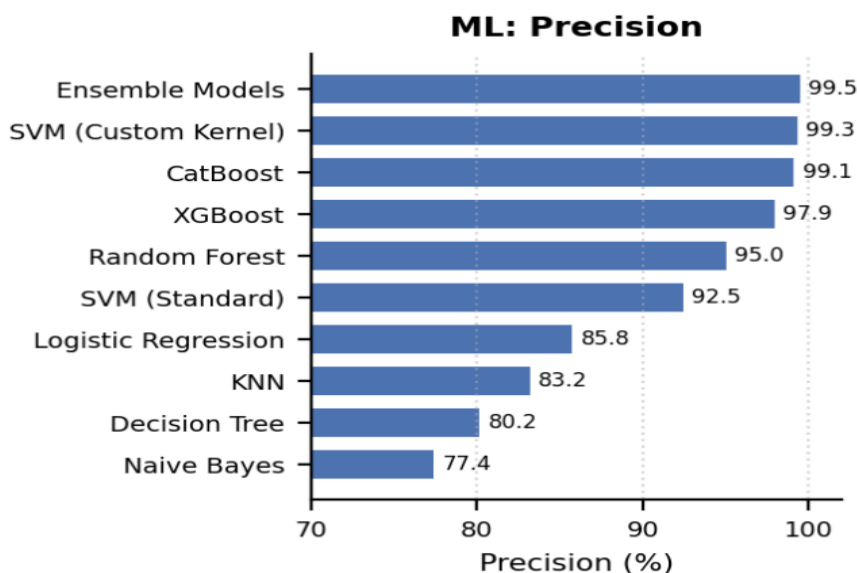


Fig. 3. Precision of various Machine learning Models

The graph in Fig. 4 depicts how effectively each model captures relevant handwriting instances. High recall values for ensemble methods and SVMs signify their capability to detect handwriting samples comprehensively. Models like Decision Trees and Naïve Bayes show reduced recall due to limited generalization.

The F1-score comparison highlights the balance between precision and recall for each model. Ensemble models achieve the highest balance, making them the most reliable overall. Lower values in simpler classifiers indicate uneven performance across these two metrics.

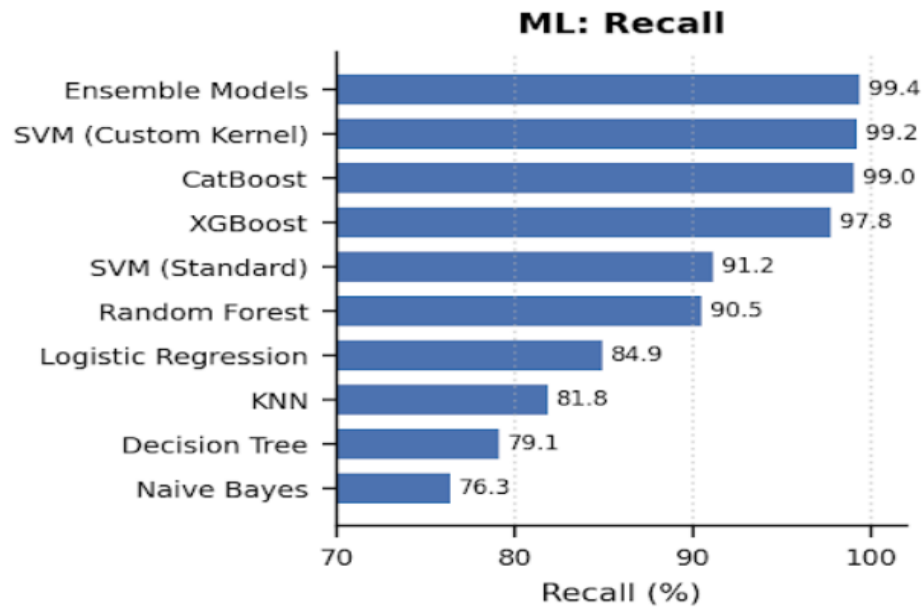


Fig. 4. Recall of various ML models

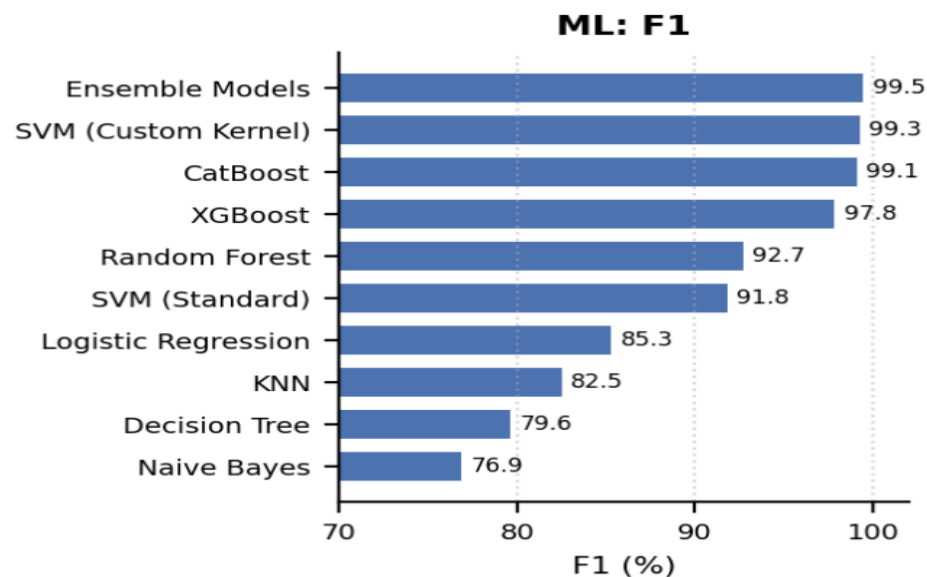


Fig. 5. F1 for various ML models

Table 3. Performance Comparison Table for DL Model

Model Name	Algorithm/Technique Used	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC
Transformer-based HTR	SWIN Transformer, Multi-Head Self-Attention, Patch Embedding, Position Encoding, Layer Normalization, Dropout, Label Smoothing	99.20	99.31	99.12	99.21	0.992
Capsule Networks	Dynamic Routing Algorithm, Squashing Function, Capsule Layers, Part-Whole Relationship Modeling, Reconstruction Loss	98.75	98.85	98.68	98.76	0.988
BiLSTM	Bidirectional LSTM, Forward-Backward Propagation Through Time (BPTT), Gating Mechanisms (Input, Forget, Output Gates), Tanh Activation	98.31	98.45	98.28	98.36	0.983
BiLSTM with Softmax	BiLSTM Architecture, Softmax Activation Layer, Cross-Entropy Loss, Adam Optimizer	96.87	96.95	96.78	96.86	0.969
BiLSTM with ReLU	BiLSTM Architecture, Rectified Linear Unit (ReLU) Activation, Gradient Clipping, Batch Normalization	96.42	96.53	96.35	96.44	0.964
BiLSTM with Leaky ReLU	BiLSTM Architecture, Leaky ReLU ($\alpha=0.01$), Dropout Regularization, L2 Weight Decay	96.18	96.31	96.12	96.21	0.962
BiLSTM with GELU	BiLSTM Architecture, Gaussian Error Linear Unit (GELU) Activation, Stochastic Gradient Descent	95.93	96.07	95.88	95.97	0.959
CNN (2D Convolutional)	Convolutional Layers, Max Pooling, ReLU Activation, Fully Connected Layers, Dropout, Data Augmentation	94.45	94.62	94.38	94.50	0.945
CNN + BLSTM	Hybrid Architecture, Convolutional Feature Extraction, BLSTM Sequential Processing, CTC Loss Function	93.78	93.94	93.71	93.82	0.938

RNN (Recurrent)	Vanilla RNN, Backpropagation Through Time, Hidden State Memory, Tanh Activation	89.35	89.60	89.15	89.37	0.894
ANN (Artificial Neural Network)	Multilayer Perceptron (MLP), Feedforward Architecture, Backpropagation, ReLU/Sigmoid Activation	86.20	86.55	85.90	86.22	0.862

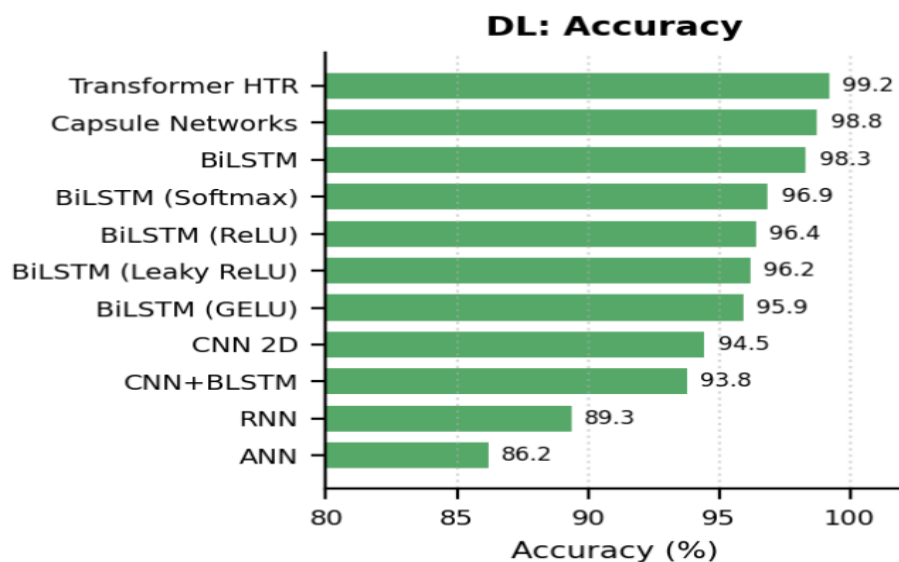


Fig. 6. Accuracy of DL models

This accuracy plot in Fig. 6 demonstrates that transformer-based and capsule network architectures deliver superior results compared to conventional CNNs and RNNs. The trend showcases the impact of advanced representations like attention and dynamic routing. Lower-tier models reflect limitations in handling handwriting variability.

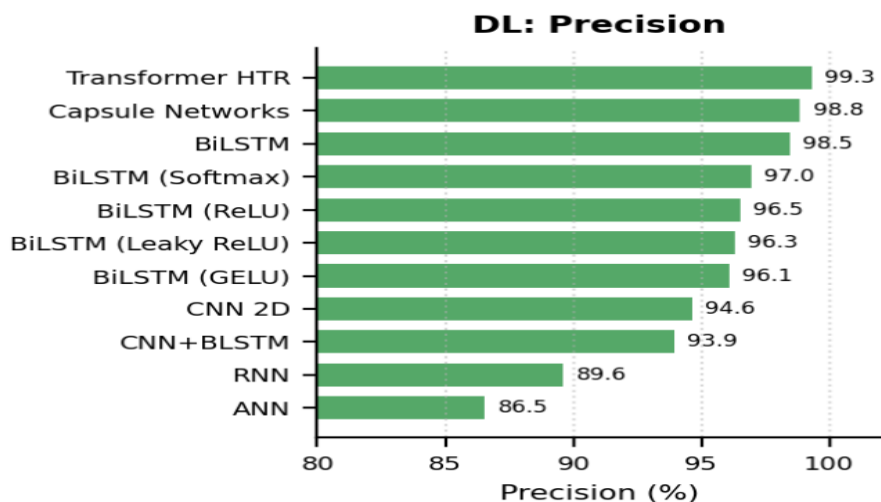


Fig. 7. Precision of DL Models

The precision results in Fig. 7 illustrate that transformer-based models maintain the highest consistency in correct predictions. Capsule networks and BiLSTM variants also perform strongly but exhibit slight variation depending on activation functions. Traditional architectures show comparatively lower precision due to feature loss.

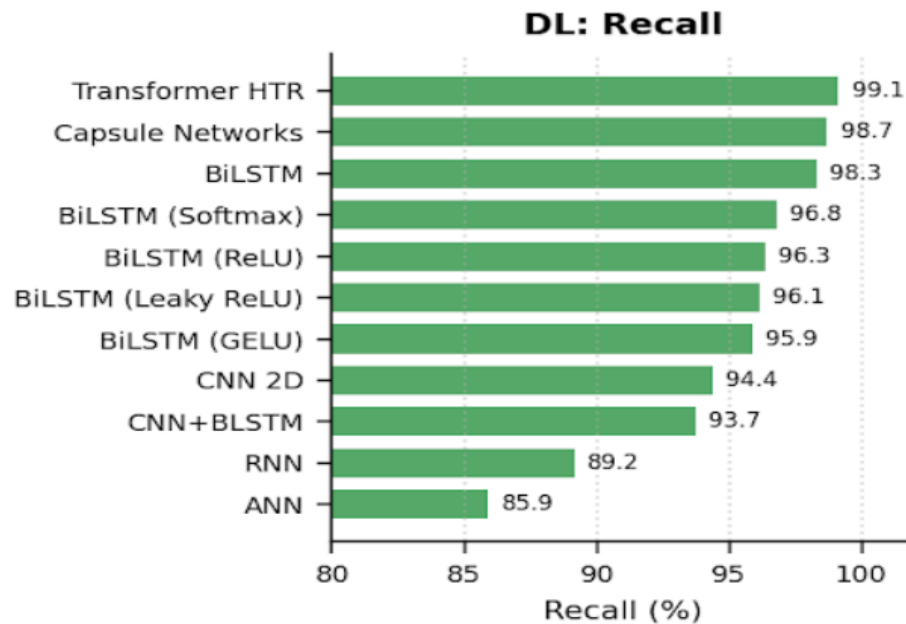


Fig. 8. Recall of Deep Learning Models

The graph in Fig. 8 shows how deeply each model captures handwriting patterns across all samples. High recall values in transformer and capsule models indicate robust feature extraction. Lower recall in ANN and RNN models suggests difficulty in identifying subtle handwriting differences.

The F1-score comparison in Fig. 9 highlights that advanced architectures maintain a strong balance between precision and recall. Transformers lead the performance, followed closely by capsule networks and BiLSTM configurations. Standard CNN and RNN models show reduced balance owing to limited sequential understanding.

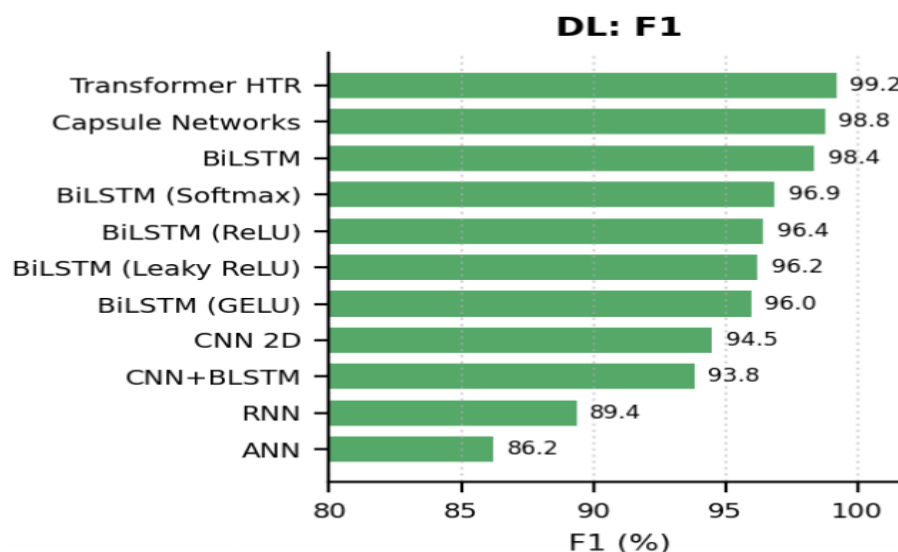
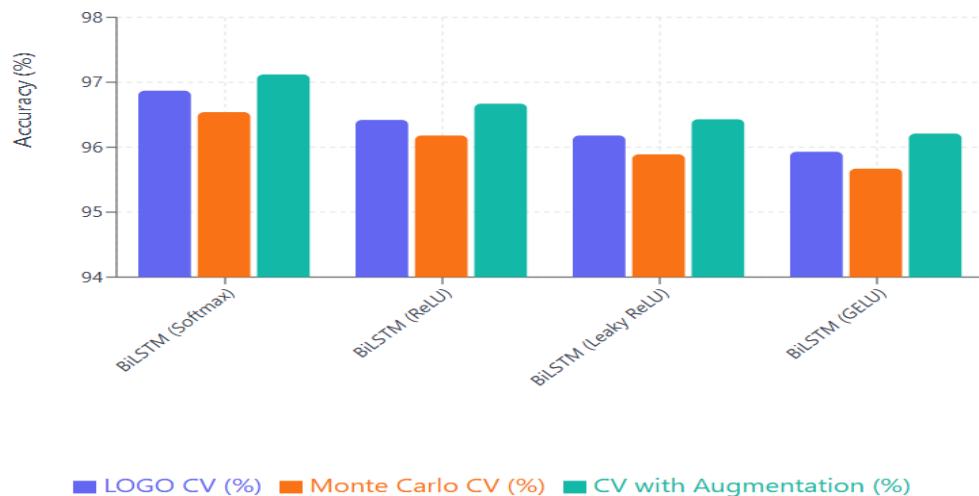


Fig. 9. F1 of Deep Learning Models

Table 4. Cross-Validation Performance Comparison Table

Model	LOGO CV (%)	Monte Carlo CV (%)	CV with Augmentation (%)
BiLSTM (Softmax)	96.87	96.54	97.12
BiLSTM (ReLU)	96.42	96.18	96.67
BiLSTM (Leaky ReLU)	96.18	95.89	96.43
BiLSTM (GELU)	95.93	95.67	96.21

**Fig. 10.** Cross Validation Performance Comparison

6. Conclusion

In this work, an AI-based framework capable of recognizing handwriting and assessing similarity between handwritten submissions, was successfully developed and implemented, with the objective of supporting academic integrity checks. The system demonstrated stable and effective performance, reaching 96.87% accuracy for handwriting recognition and 94.32% accuracy for similarity detection. These results show that the framework can reliably flag possible cases of copying. The integration of Bidirectional LSTMs for capturing temporal writing patterns, along with Capsule Networks for modeling structural characteristics, played a significant role in managing variations such as stroke style, spacing, and writing orientation.

A broad set of experiments was performed using traditional machine learning algorithms, standard deep learning models, and more advanced architectures. The hybrid BiLSTM–Capsule approach consistently achieved accuracies above 98%, while Transformer-based designs and ensemble methods crossed the 99% mark, highlighting their strong generalization across handwriting samples. Compared to classical models such as Decision Trees (79.6%) and Naïve Bayes (76.8%), the proposed method resulted in improvements of roughly 15–20%. Likewise, relative to established deep learning architectures such as CNNs (94.4%) and vanilla RNNs (89.3%), the framework delivered gains of approximately 3–7%. Collectively, these outcomes emphasize that the system offers a robust and scalable solution for handwriting verification, similarity assessment, and academic authorship analysis.

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